

Power for Non-Inferiority Trials

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Introduction

Non-inferiority trials aim to establish that a new treatment is “not meaningfully worse” than an active comparator. The non-inferiority margin is a pre-specified loss that defines the boundary. Power calculations use a one-sided test at the margin.

Prerequisites

Equivalence testing, confidence intervals.

Theory

For continuous outcomes with true difference δ (new minus reference), margin $-\Delta$ (loss), and pooled SD σ , the non-inferiority null is $\mu_{\text{new}} - \mu_{\text{ref}} \leq -\Delta$. Power at α one-sided is

$$1 - \beta = \Phi\left(\frac{\delta + \Delta}{\sigma\sqrt{2/n}} - z_{1-\alpha}\right).$$

For binary outcomes (proportions), analogous formulas using Cohen’s h or risk-difference variance apply.

Sample size increases with tighter margin, larger allowed effect, or reduced power.

Assumptions

- Pre-specified non-inferiority margin based on regulatory or clinical guidance.
- Assumed true δ (often zero, assuming equivalence).
- Normal or large-sample approximations.

R Implementation

```
library(gsDesign)

# Continuous outcome: margin -5 units, true delta = 0, sigma = 15
# One-sided alpha = 0.025, power = 0.80
delta <- 0; margin <- 5; sigma <- 15; alpha <- 0.025; beta <- 0.20
n_per_arm <- 2 * sigma^2 * (qnorm(1 - alpha) + qnorm(1 - beta))^2 / (delta + margin)^2
n_per_arm

# Binary outcome: reference rate 70% cure, margin -10%, true new = 70%
pC <- 0.70; pE <- 0.70; margin_prop <- 0.10
# Farrington-Manning or score-based calculations are standard; simplified Wald here:
p_bar <- (pC + pE) / 2
n_bin <- (qnorm(1 - alpha) + qnorm(1 - beta))^2 *
  (pC*(1-pC) + pE*(1-pE)) / (pE - pC + margin_prop)^2
n_bin
```

```
# Group-sequential non-inferiority design
ni <- gsDesign(k = 2, test.type = 1, alpha = 0.025, beta = 0.20,
              sfu = "0F")
gsBoundSummary(ni)
```

Output & Results

Continuous: $n \approx 140$ per arm. Binary: $n \approx 196$ per arm.

Interpretation

“To establish non-inferiority with a 5-unit margin, assuming a true zero difference and SD 15, at 80 % power and one-sided $\alpha = 0.025$, 140 patients per arm are required.”

Practical Tips

- Margin selection is the most important decision in non-inferiority design; regulators require explicit justification.
- Use one-sided $\alpha = 0.025$ to keep consistency with two-sided 0.05 superiority conventions.
- Report per-protocol analysis alongside intention-to-treat; dilution effects in ITT make non-inferiority easier to show artificially.
- Include interim futility stops for ethical reasons.
- Conservative margins and high power requirements inflate n quickly.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation